



on their way. Output can be seen on the browser page.

Testing procedure

To test, open a browser window in Raspi and type the following:

`http://(ip address of Raspi)/sms.php`

The window shown in Fig. 5 will appear.

Insert the message in less than 200 characters and the destination numbers separated by commas. You can copy and paste in this text box as well. Press Send and all SMSes will be sent one by one and delivery reports will appear below the browser window.

The OK in the report shows that

the SMS is delivered; if not, look for reasons. However, sometimes even when OK appears, it takes some time for the messages to get delivered.

I have put a delay of six seconds due to my network speed problems. You can change it as per your mobile network conditions.

Enter Delivery Time window will ask for delivery time (Fig. 6), which can be entered using following AT commands. If you enter mobile numbers separated by commas, an SMS will be sent to all recipients starting from the time entered with a duration of one minute between each SMS.

\$>sudo atq # will show all
scheduled tasks with task numbers
ready for delivery

\$ > sudo /var/mail/mail # will show the mail where the last delivery report available.

\$ > sudo atrm (number) # will delete any scheduled SMSes, so take help from sudo atq

Further application

SMS server with delivery schedule. An SMS server combined with AT command can make the SMS server an SMS scheduler, which can

schedule your SMSes to be sent anytime in the future starting from one minute to ten years time frame (Fig. 7).

It is simple yet extremely powerful. First, install AT command and then give execution right to www-data.

```
$> sudo apt-get install at
```

This will install AT commands on Raspi.

```
$>sudo visudo
```

Add the following line:

```
www-data ALL = (ALL) NOPASSWD: /usr/bin/
/gammu, NOPASSWD: /usr/bin/at
```

This gives permission to `www-data` as `sudo` users to run `Gammu` and `AT`.

Save visudo by pressing ctrl + o
and exit by pressing ctrl + x.

\$ > sudo gammu --sendsms EMS
your number -text "Hello guys from
EFY" is a normal SMS command
using Gammu.

```
$ > sudo gammu --sendsms EMS
your number -text "Hello guys from
EFY" | at now + 10 minutes will
send the SMS in ten minutes from
the sent time.
```

\$ > sudo gammu --sendsms EMS
your number -text "Hello guys from
EFY" | at 1430 2> &1 will send an
SMS at 2:30 PM. The '2> &1' com-
mand at the end will disable writing
this command in the log file.

```
$ > sudo gammu --sendsms EMS
your number -text "Hello guys from
EFY" | at 2:30 PM 10/21/2015 2 >&1
SMS will be sent on 10/21/2015 at
2:30 PM.
```

This can be added very easily in the PHP script (sms1.php).

Copy sms1.php file into your server's document root directory and then run from a browser window. ([http://\(your IP address\)/sms1.php](http://(your IP address)/sms1.php)) **EFY**

